

AI-Driven Interpretation of Nonverbal Communication in AR-Enhanced Real-Time Captions: Effects on Cognitive Load, Comprehension, and User Engagement

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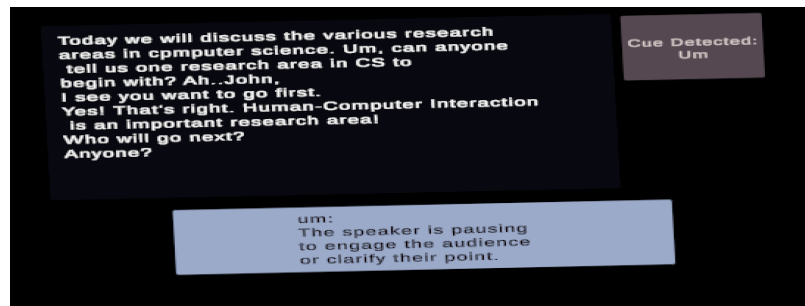


Figure 1: Proposed System: An AR real-time captioning system that uses AI to detect multimodal cues and interpret them in real time.

Abstract

Current real-time captioning systems focus on transcribing speech, often overlooking facial expressions, body language, and vocal prosody that convey essential communicative cues. We present an AI-driven augmented reality (AR) captioning system that interprets non-verbal signals in real time and renders them as dynamic visual cues within the user's view. Grounded in Cognitive Load Theory, cross-modal plasticity, and computational creativity, our approach supports Deaf and Hard of Hearing (DHH) and neurodiverse learners by transforming captions into creative, expressive media. We explore: (RQ1) how non-verbal cues affect comprehension, engagement, and creative interpretation; (RQ2) how cultural differences influence cue perception; and (RQ3) what AI and design strategies enable low-latency, customizable AR captions without increasing cognitive load. A user study shows 45% comprehension gains and 25% reduction in mental demand with emotional indicators in captions. Future work includes building a cross-cultural cue corpus, an open-source AR captioning pipeline, and design guidelines for inclusive STEM education, advancing accessibility and fostering creativity-driven communication.

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CCS Concepts

• **Human-centered computing** → **Accessibility design and evaluation methods.**

Keywords

Accessibility, Augmented Reality, Non-Verbal Communication, AI-Driven Interpretation, Deaf and Hard-of-Hearing, Cross-Cultural Design, STEM Education

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1 Introduction

Spoken language is multimodal: facial micro-expressions, body posture, gesture, and vocal prosody complement lexical content and convey affect, emphasis, and cue [18]. Conventional automatic speech recognition (ASR) captions strip away this non-verbal layer, forcing Deaf and Hard-of-Hearing (DHH) students to *choose* between reading text and watching the speaker [6]. The cognitive cost is particularly acute in STEM lectures, where dense terminology, rapid concept shifts, and visual demonstrations abound [3]. Our system transforms captions from static transcripts into creative communicative artefacts, inviting users to interpret subtle emotional cues, fostering richer, more imaginative forms of engagement in STEM learning.

This doctoral research treats captions not as inert transcripts but as *creative, expressive media*. By fusing ASR, speech-emotion recognition (SER), facial-expression analysis, and AR head-mounted displays (HMDs), we aim to encode non-verbal cues inside the caption stream, reduce split attention [12], and empower DHH learners to participate equitably. The work aligns directly with the **Creativity for Change** theme of C&C 2025: it leverages computational creativity to transform a mundane accessibility aid into an interactive artefact that reshapes classroom power dynamics.

2 Research Objectives and Questions

Aim. Design, implement, and evaluate an AI-enhanced AR captioning ecosystem that interprets non-verbal communication in real time and visualises it in cognitively efficient, culturally intelligible ways.

Primary Research Questions.

RQ1 How does the integration of facial expressions, text highlighting, emoji, or other emotional cues in real-time STEM captions affect comprehension, cognitive load, and learners' creative interpretation of emotional communication for DHH and neurodiverse students?

RQ2 How do cultural differences influence the interpretation of non-verbal cues by DHH and neurodiverse students, and what are the implications for designing culturally adaptive captioning systems?

RQ3 What are the key challenges, needs, and opportunities for DHH and neurodiverse users when interacting with LLMs and AR for real-time non-verbal communication, and how can these inform the design of an AI-enhanced AR system?

3 Background and Related Work

Cross-modal plasticity research shows that DHH individuals develop heightened peripheral and motion sensitivity [2], suggesting opportunities for spatially distributed cues in AR environments. **Cognitive Load Theory (CLT)** further posits that well-aligned multimodal signals can lower extraneous load and enhance germane processing [15], reinforcing the importance of integrating non-verbal signals into captioning systems. Recent projects such as *Visible Nuance* [9] and *Caption Royale* [4] have begun to visualize paralinguistic features in captions, but primarily target entertainment media and do not address the challenges of real-time latency, cultural variation, or AR-based delivery for educational contexts.

Building on these foundations, our work advances the state of the art by uniting CLT, cultural theory, and AR technology to deliver adaptive, real-time captions in live STEM classrooms. Furthermore, it draws on emerging ideas in computational creativity [10], exploring how AI-mediated captioning can not only enhance accessibility but also stimulate imaginative engagement and novel interpretation strategies among users.

4 Work Already Done

4.1 Completed Study 1: Emotion Enhanced Captions

Primary RQ1: How does the integration of facial expressions, text highlighting, emoji, or other emotional cues in real-time STEM

captions affect comprehension, cognitive load, and learners' creative interpretation of emotional communication for DHH and neurodiverse students?

RQ1b: In what ways do different approaches to visualizing emotional and prosodic information (e.g., on-screen emoji, overlay on the speaker video, textual emphasis) shape user preferences, perceived usability, and perceived connection with the instructor?

Our submitted ACII 2025 paper "*Adaptive Captioning with Emotional Cues: Supporting DHH and Neurodiverse Learners in STEM*" (under review) explored four prototype designs (Fig. 2). We developed a web-based captioning system that embeds emotional cues into STEM lectures, aiming to highlight nonverbal aspects (e.g., vocal emphasis, instructor gestures) for learners who are DHH or neurodiverse.

Implemented using HTML, CSS, and JavaScript, the model processes a pre-recorded ~4-minute lecture, segments the transcript into individual words, and annotates them with affective attributes such as color, animation, or icons.

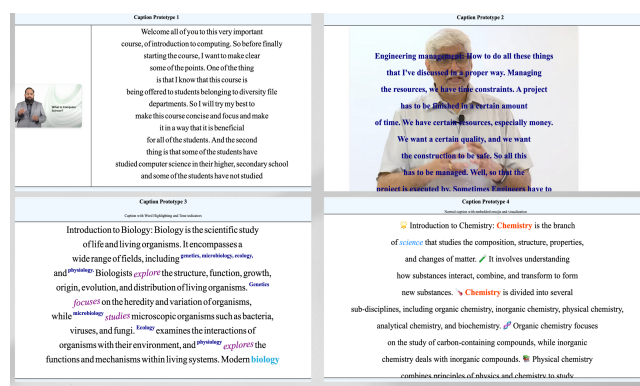


Figure 2: Four web-based captioning prototypes designed to enhance emotion in STEM lectures: (P1) plain text beside speaker video, (P2) text overlaid on video, (P3) highlighted text with animations, and (P4) captions with emojis and discipline-specific icons.

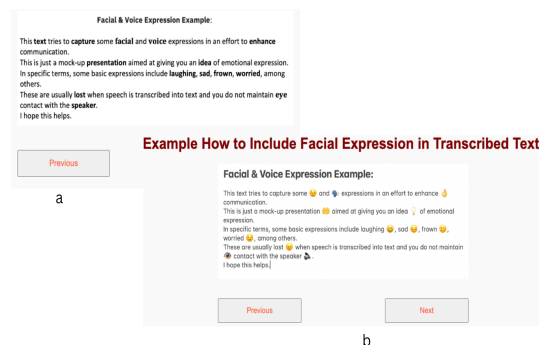


Figure 3: Proposed caption design mock-up

Each prototype used the same underlying transcript but was paired with a different STEM domain video to minimize memorization effects and examine adaptability across topics:

- **P1: Text + Speaker Video** — Captions appear beside the instructor’s video, preserving access to facial expressions and gestures but without added visual emphasis.
- **P2: Caption Overlay** — Captions are overlaid on the video to reduce eye movement, though potentially occluding facial cues critical for lipreading.
- **P3: Highlighted Text + Animation** — Key terms are bolded or color-highlighted with subtle animations to convey prosody, avoiding use of emojis.
- **P4: Icons/Emojis + Discipline Visuals** — Builds on P3 by adding expressive emojis and subject-specific icons (e.g., lab flasks), enhancing emotional cues while increasing visual complexity.

Though domains varied (e.g., Engineering for P2, Biology for P3, Chemistry for P4), this approach enabled us to assess how emotional captioning can support comprehension across different technical contexts.

4.2 Method

A pilot (n=5) informed refinements, and the study procedure involved participants first watching a short recorded STEM lecture (Chemistry, 3 minutes) with only raw ASR transcripts.

They then reviewed design mockups (Figure 3), illustrating possible enhancements: (i) bold or colored highlighting of key/emotion-laden words, and (ii) emojis inserted to convey speaker mood (Figure 3). We collected open-ended feedback via an online survey (QuestionPro).

The main study involved 24 participants (8 DHH, 7 ADHD, 1 dyslexia). Each watched four short STEM clips of the web-based prototypes described in (Figure 2, after that they answered comprehension questions, and completed NASA-TLX.

The participant cohort included key populations of interest: 33.3% (n=8) identified as Deaf or Hard of Hearing (5 Deaf, 3 Hard of Hearing), while 29.2% (n=7) reported ADHD and 4.2% (n=1) dyslexia. Caption usage patterns revealed 92% were regular users (split evenly between always and frequent users), with 58.3% rating captions as “extremely useful” for their needs.

4.3 Key Findings

Comprehension: Correct answers rose from 54% (baseline captions) to nearly 100% with emotion enhanced designs. An independent *t*-test showed that these differences were statistically significant ($p < .001$), albeit confounded by the fact that different lecture content was used.

Table 1: Repeated-measures ANOVA results for NASA-TLX subscales. Values in parentheses represent partial η^2 .

Subscale	F-value	p-value	partial η^2
Mental Demand	$F(3, 10824.458) = 4.6065$	$p = 0.0060$	0.0013
Physical Demand	$F(3, 9221.0348) = 3.4898$	$p = 0.0216$	0.0011
Effort	$F(3, 8279.4334) = 3.6820$	$p = 0.0173$	0.0013
Frustration	$F(3, 20878.496) = 6.7620$	$p = 0.0006$	0.0010

Cognitive Load: Prototype 3 (keyword highlighting) reduced mental demand by 25% compared with overlay captions ($F(3, 10824.458) = 4.61$, $p = .006$).

A repeated-measures ANOVA (Table 1) revealed significant effects of prototype on *Mental Demand*, *Physical Demand*, *Effort*, and *Frustration* (all $p < 0.05$). Despite reaching statistical significance, effect sizes were small (partial $\eta^2 \approx 0.001$), likely reflecting the large error term in our model. Specifically for Mental Demand, the effect was:

$$F(3, 10824.458) = 4.6065 \quad p = 0.0060 \quad \text{partial } \eta^2 = 0.0013$$

Post-hoc comparisons confirmed that P2 was significantly more demanding than P3 ($p < 0.05$). Overall, these findings suggest that P2’s overlaid design may impose additional visual load, while the minimal, keyword-highlight approach in P3 was comparatively less mentally taxing.

User Preferences: DHH participants favoured designs that balanced expressiveness and legibility; neurodiverse users valued customisation toggles. Emoji were polarising—useful to some, distracting to others—underscoring the need for personalisation, summarized in (Figure 4).

Participants engaged not only in passive reception but also in active, creative decoding of emotional subtleties embedded in captions, demonstrating the interpretive richness enabled by our designs.

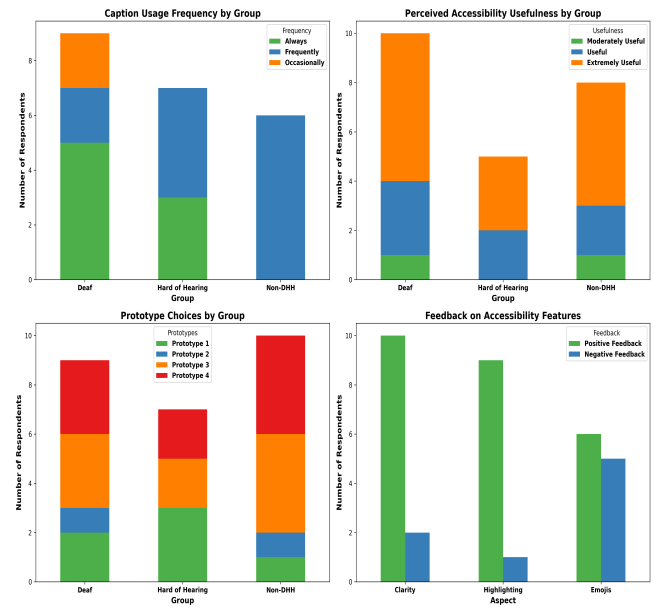


Figure 4: Comparison of caption usage frequency, perceived usefulness, prototype preferences, and feedback on accessibility features across Deaf, Hard of Hearing, and neurodiverse respondents.

4.4 Summary of Research Questions Addressed

Study 1 addressed Primary RQ1 by demonstrating that emotion-enhanced caption designs improved comprehension and reduced cognitive load for DHH and neurodiverse learners, with minimal

designs such as keyword highlighting yielding the best cognitive outcomes. Furthermore, it partially addressed creative interpretation, as participants engaged in active decoding of emotional subtleties embedded in the captions. The study also addressed Secondary RQ1b by showing that different visualizations (e.g., text highlighting, emojis, video overlays) shaped user preferences and perceived usability, with findings highlighting a tension between expressiveness and clarity, and revealing a strong desire for customizable captioning features.

4.5 Implications

Results confirm that subtle visualisation of non-verbal cues can enhance learning while excessive embellishment risks overload. They also reveal cultural and individual variability, motivating RQ2 and the adaptive, user-configurable approach planned for Study 3.

5 Study 2 – Exploring Cultural Differences in Non-Verbal Cue Interpretation (RQ2)

5.1 Motivation

Cultural differences play a significant role in how non-verbal cues are interpreted, which is crucial for developing inclusive technologies. Research shows that non-verbal communication is not universal; it is deeply embedded in cultural norms and practices [7, 8]. For example, facial expressions or gestures may convey curiosity in one culture and skepticism in another [5, 11]. This presents challenges for DHH and neurodiverse individuals who rely heavily on captions and visual signals for communication, as existing captioning systems often overlook such cultural variations [1].

5.2 Theoretical Framework

This study draws on **Hofstede’s cultural dimensions theory** [8] and **Hall’s high-context versus low-context communication framework** [7] to explain differences in non-verbal cue interpretation. These theories help analyze how cultural factors such as individualism versus collectivism and context-dependence influence perception. In addition, **Ekman’s theory of universal emotions** [5] serves as a baseline to identify culture-specific display rules, offering a foundation for cross-cultural comparison.

5.3 Research Questions

Primary RQ2: How do cultural differences influence the interpretation of non-verbal cues by DHH and neurodiverse students, and what are the implications for designing culturally adaptive captioning systems?

- (1) **RQ2a.** What universal and culture-specific patterns exist in how DHH and neurodiverse students from diverse backgrounds interpret non-verbal cues?
- (2) **RQ2b.** How do cultural factors (e.g., individualism vs. collectivism, high-context vs. low-context communication) shape these interpretations?
- (3) **RQ2c.** What design principles can be derived from these insights to inform culturally adaptive captioning systems?

5.4 Methodology

Phase 1 – Exploratory Study (RQ2a and RQ2b). The goal of this phase is to identify universal and culture-specific patterns in the interpretation of non-verbal cues among DHH and neurodiverse students. Semi-structured interviews will be conducted with students in the United States, focusing on participants whose cultural backgrounds align with high-context (e.g., East Asian, Middle Eastern heritage) and low-context (e.g., North American/Western European heritage) communication styles. Participants will engage in design workshop scenarios showcasing varied non-verbal cues (e.g., facial expressions, gestures, body language) embedded in culturally meaningful contexts. Think-aloud protocols and follow-up interviews will be used to capture interpretation strategies and the influence of cultural communication styles.

Phase 2 – Theoretical Analysis (RQ2b). Qualitative data from Phase 1 will be thematically coded using NVivo and deductively mapped onto Hofstede’s and Hall’s theoretical frameworks. This phase will identify specific cultural dimensions that influence interpretation, ensuring inter-coder reliability with Cohen’s kappa $\kappa \geq 0.80$. The outcome will be a theoretical model linking cultural factors to patterns of cue interpretation.

Phase 3 – Design Implications (RQ2c). Two co-design workshops per culture (with approximately eight participants each) will be conducted using Miro or Figma to remix caption styles. Participants will collaboratively ideate on visual grammar, placement, and contextual cues that enhance comprehension. These insights will be synthesized into design heuristics and contribute to a shared *Cross-Cultural Cue Corpus*.

5.5 Data Analysis

Quantitative analysis (e.g., frequency counts, chi-square tests) will be used to measure patterns in cue interpretation. Qualitative data will be analyzed using grounded theory coding and cross-case synthesis. Joint display matrices will align statistical findings with thematic insights, supporting the derivation of actionable design principles.

5.6 Expected Outcomes

- (1) **Cue Interpretation Taxonomy:** Categorization of universal and culture-bound interpretations of non-verbal cues.
- (2) **Predictive Cultural Model:** Framework linking cultural dimensions to decoding strategies used by DHH students.
- (3) **Design Heuristics:** Guidelines for creating culturally adaptive captions (e.g., emphasizing peripheral animations in high-context cultures).
- (4) **Public Dataset:** A curated, anonymized dataset to support replication and further machine learning research in inclusive captioning.

6 Study 3 – AI Enhanced AR Captioning System

6.1 Motivation

Non-verbal cues like facial expressions, gestures, and vocal tones play a vital role in providing context that improves understanding and emotional connection [5, 13]. Yet, current systems face challenges in accurately and culturally adapting these cues [1].

Further, the choice of AR for this study is based on several reasons: first, to minimize the split-attention effect where users switch gaze between their system and the instructor [12, 17], second, the finding that in an inclusive classroom, displaying captions on a classroom projector can potentially distract other students, similar to how not everyone in a group setting watching television may want captions on, and third, using AR captions can provide a layer of privacy for DHH and neurodiverse students who prefer not to disclose their disability. When captions are personalized and discrete, it minimizes awareness among peers, thus avoiding unwanted attention. Supporting this, Seita et al. [14] found that hearing individuals tend to alter their speaking behaviors—such as modifying speech rate, intensity, and gestures—when they know their speech is being transcribed. This change can disrupt the natural communication flow, making interactions feel less authentic. AR captions provide an alternative that respects both the user's privacy and the group's dynamics, facilitating a more natural experience for all participants [16].

6.2 Theoretical Framework

Our system design is informed by several theoretical frameworks. **Cognitive Load Theory** guides interface decisions aimed at minimizing extraneous cognitive load, such as avoiding overlaid text that occludes lip movements. **Mayer's Multimedia Principles**—including contiguity, signaling, and redundancy—further shape the multimodal alignment and content presentation strategies. Additionally, **Cross-Modal Plasticity** [2] suggests that DHH learners exhibit heightened peripheral visual attention, supporting the spatial distribution of cues within AR environments. Finally, findings from Study 2 provide culture-specific parameters for visual grammar and interaction design, ensuring that the system is sensitive to diverse communication norms.

6.3 Research Questions

Primary RQ3: What are the key challenges, needs, and opportunities for DHH and neurodiverse users when interacting with LLMs and AR for real-time non-verbal communication, and how can these inform the design of an AI-enhanced AR system?

- (1) **RQ3a.** What are the current challenges DHH and neurodiverse users face when using LLMs and AR for real-time communication, and how do they work around these challenges?
- (2) **RQ3b.** How do DHH and neurodiverse users envision using LLMs and AR to enhance non-verbal communication in real-time contexts?
- (3) **RQ3c.** What design principles and features are most effective for addressing the identified challenges and meeting user needs in an AI-enhanced AR system?

6.4 Methodology

Phase 1: Exploratory Study (RQ3a and RQ3b). Objective: To gain insights into the challenges faced by DHH users when using LLMs and AR, along with their needs and expectations for real-time non-verbal communication.

Methods:

Conduct semi-structured interviews with DHH users to investigate:

Their current experiences with LLMs and AR.

The challenges they encounter and the workarounds they employ in real-time communication.

The features and functionalities they desire in an AI-enhanced AR system.

Utilize low-fidelity mockups or prototypes to collect feedback on potential designs.

Outcome: A comprehensive understanding of user needs, challenges, and design opportunities that will guide the development of the prototype.

Phase 2: Prototype Development (RQ3c). Design an AI-enhanced AR system that integrates user feedback and addresses the challenges identified, then employ iterative design and testing with users to refine the prototype.

Outcome: A validated prototype that meets user needs and adheres to design principles.

Phase 3: Evaluation Study. To assess the effectiveness of the prototype in improving real-time non-verbal communication for DHH users.

Methods:

- Conduct a usability study with DHH participants using the prototype.
- Evaluate usability (SUS), cognitive load (NASA-TLX), and comprehension (test scores) under various conditions (e.g., simple vs. complex content, standard vs. enhanced captions).
- Collect qualitative feedback on user experiences and perceived benefits.

Outcome: A set of design guidelines and validated insights into how AI and AR can improve non-verbal communication for DHH users.

6.5 Timeline

The proposed timeline outlines the key milestones from mid-stage research activities through to dissertation completion. With the preliminary exam successfully completed and the first manuscript submitted to ACII 2025, the project now moves into the implementation and evaluation phases. Summer 2025 will focus on developing and conducting the user study for Research Question 2 (RQ2), with the goal of preparing a submission to ACM ASSETS 2026. In Fall 2025, the emphasis shifts to the implementation and user study for Research Question 3 (RQ3), targeting submissions to IEEE VR 2026 and ACM CHI 2026. The final phase, Spring 2026, will be dedicated to dissertation writing, culminating in the research defense and final defense. This structured timeline ensures a steady and achievable path toward the successful completion of the dissertation.

Table 2: Proposed Timeline and Milestones

Timeframe	Milestones
Spring 2025	• Preliminary Exam - Done • Finalize Writing for RQ 1 • ACII 2025 – Submitted
Summer 2025	• Implementation of RQ 2 User Study and Prototype • Recruiting & Writing for RQ 2 User Study • ACM ASSETS 2026
Fall 2025	• Implementation of RQ 3 User Study and Prototype • Recruiting & Writing for RQ 3 User Study • IEEE VR 2026 / ACM CHI 2026
Spring 2026	• Dissertation Writing • Research Defense • Final Defense

6.6 Anticipated Contributions

- **Theory:** Extension of Cognitive Load Theory (CLT) quantifying trade-offs when non-verbal cues are algorithmically re-encoded; a model of cultural mediation in caption interpretation.
- **Technology:** First open-source AR captioning pipeline combining speech-to-text (STT), speech emotion recognition (SER), and facial action analysis, with sub-300 ms latency on consumer hardware.
- **Design Knowledge:** Culturally adaptive visual grammar for captioning; design guidelines that leverage the peripheral vision strengths of DHH users.
- **Empirical Evidence:** Multi-study dataset linking caption modality to learning outcomes across diverse cultural contexts.
- **Societal Impact:** Blueprint for privacy-preserving, creativity-driven change in inclusive STEM education.
- **Creative Engagement:** Evidence and models showing how dynamic, affect-enhanced captions can stimulate interpretive creativity and imaginative meaning-making among DHH and neurodiverse learners.

7 Conclusion

This dissertation proposes an AI-driven captioning ecosystem that reimagines accessibility not as a matter of static compliance, but as a space for dynamic, creative expression. By integrating non-verbal cues—facial expressions, gestures, and vocal tone—into real-time AR captions, the system supports deeper comprehension, reduced cognitive load, and more engaging learning experiences for DHH and neurodiverse students.

Study 1 demonstrated that emotion-enhanced captions significantly improve comprehension and reduce mental effort, affirming the cognitive benefits of multimodal signaling. Study 2 will build on these findings by investigating cultural variability in cue interpretation, establishing design principles for culturally adaptive systems. Study 3 will culminate in the development and evaluation of a real-time AR prototype that operationalizes these insights in inclusive classroom settings.

Together, these studies chart a path toward captions as expressive, interactive media that foster not only understanding but also

imagination and agency. Limitations—such as small effect sizes and the challenge of cross-cultural generalizability—will guide future work on scalable datasets, participatory design in underserved communities, and long-term classroom deployment.

Ultimately, this work contributes to a broader vision of accessible technology: one that embraces the richness of human communication and positions DHH and neurodiverse learners not as passive recipients, but as co-authors in the meaning-making process.

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